

Nonlinear Computer Model for the Simulation of Lock-in Vibration on Long-Span Bridges

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Abstract: *The susceptibility of modern bridges to vortex-shedding-induced vibration is a major concern for researchers and designers. The relevance of this phenomenon is associated with the onset of large-amplitude aeroelastic vibration at moderate wind-velocity regimes due to synchronization, that is, lock-in, of the vortex-shedding frequencies with those corresponding to the natural modes of the structure. Recent observations, either recorded during the monitoring of full-scale bridges or during experimental tests of deck models in wind tunnels, confirm the importance of these aspects during the operational life of the structure. In this article, a computer model for the simulation of the aeroelastic loading associated with vortex shedding in lock-in regime is presented, for a direct application to dynamic analysis of long-span bridges. This approach is based on earlier work focused on the response of slender vertical cylindrical chimneys to vortex-shedding excitation, which is here extended to noncircular cross sections. The numerical model was employed in conjunction with a finite-*

element code for time-domain nonlinear simulation of the structural dynamic response. The validation of the procedure is performed through numerical simulation, conducted on two specific bridge examples.

1 INTRODUCTION

The aerodynamic and aeroelastic wind-induced actions on a long-span suspension bridge can be attributed to many distinct phenomena, not only affecting the overall structure but also specific portions, in accordance with apparently different criteria.

First, aerodynamic actions on the deck (usually corresponding to the surface mainly exposed to wind) are considered. In this case, the aerodynamic analysis is based on the evaluation of stationary force coefficients under mean uniform wind (static coefficients), measured through wind tunnel tests on reduced-scale models of the deck.

The assessment of the flutter instability threshold (critical wind speed) can be successfully performed

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by means of the nonstationary frequency-dependent aeroelastic derivatives (Simiu and Scanlan, 1996) or analogous time-evolution coefficients, such as the Modified Quasi-Steady model proposed by Diana et al. (1995), and the Indicial Function approach (e.g., Chen et al., 2000; Borri and Höffer, 2000; Scanlan, 2000; Caracoglia and Jones, 2003). These nonstationary forces are correlated with the static coefficients only for low-frequency oscillations and include in the more general case the unsteady contribution due to the modifications on the leeward wake due to the deck oscillation (down-wash effect). The corresponding critical wind speed is obtained by means of models with two or more degrees of freedom (e.g., Simiu and Scanlan, 1996; Jain et al., 1996; Katsuchi et al., 1999; Sepe et al., 2000).

Several authors have recently highlighted the relevance of the effects associated with vortex shedding from the bridge deck for wind speed in the lock-in interval. Vibrations have been in fact observed on real structures either during construction or after the opening of the structure (Scanlan, 1998; Kawatani et al., 1999; Larsen and Jacobsen, 1992; Lee et al., 1997). An example from the past is represented by the Menai Strait Suspension Bridge in England (1826), corresponding to the first recorded event of vortex-induced vibration observed during the erection stages of the structure (Buonopane and Billington, 1993).

In 1997, the Storebaelt suspension bridge in Denmark experienced low-frequency vertical oscillation during the final stage of deck erections for wind speeds between 4 and 12 m/s (Larsen et al., 2000; Frandsen, 2001); on-site monitoring of the vibration and wind speed measurements indicated that the cause of the vibration was periodic vortex shedding; specific countermeasures were adopted, including the design of appropriate guide-vanes.

In addition, vortex-induced steady-state vibration has been observed on other categories of bridges, characterized by extreme slenderness of the deck. An interesting example, recently documented, is related to the multi-span bridge of the Trans-Tokyo Bay Highway Crossing. The two longest spans of the bridges, completed in 1997, experienced significant vibration due to vortex shedding under prevailing winds orthogonal to bridge axis, with wind velocity around 16–17 m/s, resulting in maximum amplitudes exceeding 50 cm (Fujino and Yoshida, 2002). Another example can be derived from the full-scale long-term monitoring of the Fred Hartman Bridge, a twin-deck cable-stayed bridge crossing the Houston Ship Channel between Baytown and La Porte, Texas (USA) (Ozkan et al., 2001; Mashnad et al., 2005; Liu et al., 2005). In particular, clear indication of vortex-induced oscillation was observed during a specific wind event, characterized by moder-

ately low wind velocity and low turbulence but leading to large vertical vibration amplitude of the deck dominated by one of the lower vertical modes (Liu et al., 2005). Other examples include the Rio-Niteroi bridge, spanning across the Guanabara Bay in Rio de Janeiro (Battista and Pfeil, 2000) and the Ohshima steel box-girder bridge (Yoshimura et al., 2001). Examples from the recent past also include the Kessock Bridge in Scotland, United Kingdom (Owen et al., 1996) and the Saint-Nazaire Bridge in France (Bietry et al., 1995; Szechenyi, 1973).

The key aspect of all these examples is the fact that significant vertical displacements of the deck can be noticed at relatively low wind speed (15–25 m/s), considerably lower than the flutter instability threshold and likely to occur very often during the lifetime of the structure; cyclical wind-induced oscillations might potentially cause fatigue problems on the structural elements (e.g., hangers, in particular at the anchorages) and the reduction of travel comfort levels for both road and railroad users.

In such systems special attention is sometimes paid to problems exclusively evident during the erection stages of the bridge (Brancaleoni, 1992), or to the behavior of selected structural elements, such as the vibration of vertical hangers due to vortex shedding and interference galloping effects due to the presence of closely spaced cables (e.g., Tokoro et al., 2000). In any case, the assessment of the aeroelastic and aerodynamic response of a suspension bridge requires computational techniques more and more sophisticated as the deck span increases. The structural analysis of such a complex system is often conditioned by the large number of degrees of freedom that is required for an accurate modeling of the response.

For long-span suspension bridges (main span variable between 3,000 and 4,000 m), a new class of wind-induced dynamic phenomena becomes of relevance. For this category of bridges critical coupling between vertical and torsional modes typically involves frequency intervals between 0.05 and 0.07 Hz (period T variable between 15 and 20 seconds), usually connected to either the risk of global deck vibration under turbulent-wind conditions and, ultimately, the collapse due to flutter-type instability (e.g., Simiu and Scanlan, 1996; Katsuchi et al., 1999). Nevertheless, at the same time the diameter of the main cables may be large (between 1 and 2 m) and therefore aerodynamic (and vortex-shedding) loading and subsequent effects cannot be *a priori* considered as negligible. This observation is also corroborated by recent studies (Diana et al., 2006), in which the importance of vortex-shedding-induced vibration on box-type bridges was highlighted during the wind-tunnel aerodynamic design of the

proposed suspension bridge for the Messina Strait in Italy.

Therefore, a more detailed analytical and numerical study of such effects is necessary, also considering potential interactions and amplification effects with other local or global phenomena. In particular, the combination of these phenomena cannot be possibly ruled out, when the characteristic frequency intervals tend to become relatively close.

Computer models for the dynamic analysis of bridges have become a research and design tool for most applications in bridge engineering. However, the simulation and implementation of the wind loading in the presence of nonlinear aeroelastic phenomena such as those related to vortex shedding in particular, are usually developed for the solution of specific problems (i.e., a particular bridge example), while the necessity of a generalization of the procedures is of paramount importance for the development of future simulation codes.

This article focuses on the discussion, understanding, and interpretation of a computer model, specifically devoted to these issues, and of some numerical simulation results, highlighting the relevance of vortex-shedding-induced aeroelastic vibration on long-span bridges in the context of nonlinear finite-element modeling for dynamic analyses.

An existing finite-element computer model (“Tenso”), developed for the assessment of aeroelastic instability (Sepe et al., 2000; Caracoglia, 2001), was used in this study.

A new procedure for wind loading and response simulation on long-span bridges was developed (coined as Global Aerodynamic Aeroelastic Model—GAAM). This computer model can simultaneously account for aerodynamic and aeroelastic forces on all structural elements of the bridge: deck, cables, and towers. In particular, the new model is capable of simulating the bridge response under the simultaneous combination of aerodynamic loading (e.g., associated with the presence of turbulence in the oncoming wind) and vortex shedding. To the authors’ knowledge, this is an original and novel aspect of the proposed model.

In this study, vortex-shedding-induced oscillation is simulated through a nonlinear model, originally proposed by D’Asdia and Noè (1998) for circular cross sections only. In this study, the model for vortex-shedding loading was re-formulated and adapted to enable the study of wind-induced effects on bridges. While the original model, developed for circular cross sections, was conceived for an application to slender chimneys, for which the assumption of linear dynamic response under vortex-shedding excitation is valid within the range of moderate vibration regimes, this can no longer be accepted in the case of long-span suspension bridges.

In fact, specific features of such structures call for an accurate representation of the nonlinear (geometric) stiffness characteristics at all regimes of vibration. Therefore, the combination of the model for vortex shedding with the nonlinear “Tenso” finite-element model, also represents an original aspect of the proposed approach.

Indeed the algorithm described herein enables for the first time the direct simulation of the response in the time domain of a combination of all relevant fluid-structure phenomena on long-span bridges. Furthermore, the proposed algorithm requires very few fluid-dynamics parameters to be specified *a priori* (such as wind velocity, basic aerodynamic force coefficients), readily available from section-model tests in a wind tunnel. In contrast, most physical quantities of the structure that directly influence the onset of the vibration (such as the modal frequencies of the mode that predominantly participate in the vibration) are detected by the computer procedure during the simulations.

Numerical simulations were conducted to validate the proposed method; namely, two simulation examples are considered, in which global and localized oscillations of the bridge and different lock-in conditions are investigated.

The article is organized as follows. In Section 2, an overview of the models for wind excitation (both vortex shedding and other aerodynamic loads) is presented along with its extension to bridges. In Section 3, the description of the computer model for finite-element dynamic simulations is described. In Section 4, examples of application are presented. Discussion on the effects of vortex shedding originating from the main cables is analyzed in Section 5, and concluding remarks are summarized in Section 6.

2 DESCRIPTION OF THE AERODYNAMIC MODELS FOR WIND EXCITATION

2.1 The model for vortex-shedding excitation

The across-wind force due to vortex shedding from a fixed cylindrical body can be described as a sinusoidal force per unit length, with frequency f proportional to the cross-flow velocity V_w . The Strouhal number $St = fH/V_w$ is a constant depending on the shape of the cross-section (Simiu and Scanlan, 1996), where the deck across-wind dimension is denoted by H . The total force value on a finite length of the cylinder depends on its longitudinal correlation and it is therefore influenced by the turbulence of the oncoming wind. The same observation applies to an elastically suspended cylinder, if the shedding frequencies are well separated from the eigenfrequencies and, consequently, the oscillation

amplitude due to the across-wind force is sufficiently small.

When the shedding frequency is close to one of the system eigenfrequencies and for low structural damping, an aeroelastic interaction between the vortex-induced force and the structural response can be observed, leading to the synchronization of the shedding to the oscillation frequency of the system (lock-in) with corresponding large-amplitude vibration of the system. The magnitude of the across-wind force and the range of the lock-in wind velocity can be related to some characteristic factors, depending on the structural response (oscillation amplitude and frequency) and, therefore, on the mechanical and aerodynamic characteristics of the deck.

The lock-in phenomenon typically shows a hysteretic and self-limiting behavior, that is, the oscillation amplitude never exceeds values of the order of the transverse dimension of the section, also for very low structural damping. A comprehensive fluid-dynamic model is not yet available in the literature for an application to a wide-range of cases, although many models have been presented but only tested on a restricted number of bluff sections (Simiu and Scanlan, 1996; Scanlan, 1998; Goswami, 1991; Goswami et al., 1992; Hansen, 1998; Ruscheweyh, 1994). However, all these models require a preliminary evaluation of some aspects of the response itself, for example, the limiting amplitude of oscillation. Their ability to forecast the maximum amplitude of the response is therefore subjected to the correct assessment of the parameters and the adequate description of the aeroelastic interaction in the entire range of wind velocities. As a consequence, the evaluation of the structural response obtained through different approaches, based on different theoretical models, may be different, in particular for weakly damped systems.

The numerical simulations described in this article were carried out by means of a model for vortex-shedding-induced loads, originally proposed by D'Asdia and Noè (1998) for circular chimneys, and is here extended for the first time to bridge deck sections.

In contrast with the model, originally developed for chimneys, and in which dynamic response of slender mono-dimensional structures with circular cross section was evaluated through a standard finite-element discretization in terms of linear beam elements, the proposed computer model allows for both structural and aerodynamic nonlinearities to be considered at the same time during the simulations. This is certainly another original feature of the computer model described in this article.

Therefore, once implemented within the structural computer model described in Section 3, the proposed

model is able to provide a reasonably correct response of the structure due to vortex shedding, including the steady-state oscillation amplitude (limit cycle). In contrast with other vortex-shedding models, available in the literature, in which nonlinear aerodynamic state equations must be added to the set of structural dynamics equations to simulate the effects of vortex shedding (calling also for the definition and calibration of a set of aerodynamic parameters that are perhaps difficult to extract from conventional wind tunnel tests), this procedure exclusively requires the initial definition of standard parameters of the deck, usually available from section-model wind-tunnel tests; namely, the model requires only the definition of Strouhal number St , drag coefficient C_D and vortex-shedding coefficient C_{L-she} , see Equations (2) to (8) below, and an estimate of the parameters Ω_L and Ω_U , defining the lock-in frequency interval. Moreover, as shown in D'Asdia et al. (2003), due to the intrinsic stability of the proposed method, also an approximate assessment of the above-mentioned parameters (e.g., derived from the literature on similar sections) is capable of accurately capturing the system response.

The time-domain simulation model is based on the hypothesis that the response of an elastically suspended body, caused by vortex shedding, can be treated as a nonlinear oscillation phenomenon.

The aerodynamic load due to the variation of the surface pressures induced by the detachment of the vortices, is defined through an alternating force per unit length, with variable direction, frequency, and phase, as a function of the relative wind velocity. The direction of the force is always perpendicular to the relative velocity.

The lock-in phenomenon is simulated by imposing an instantaneous correspondence between excitation and oscillation frequencies in a pre-selected range of values. Outside this interval, the frequency of the pulsating force is calculated according to the Strouhal Law (Simiu and Scanlan, 1996), as a function of the relative wind velocity. As the frequency varies, the continuity of the exciting force is ensured by a suitable modification of its phase.

The forces acting on the cross section are represented in Figure 1. The term Δt represents the integration time-step; the total force per unit-length due to wind action at time t for the cross section located at x (longitudinal abscissa along bridge axis) is:

$$\vec{F}_{w(t),x} = \vec{F}_{she(t),x} + \vec{F}_{drag(t),x} \quad (1)$$

The component due to vortex shedding at time t is the force $\vec{F}_{she(t),x}$, assumed as perpendicular to the relative

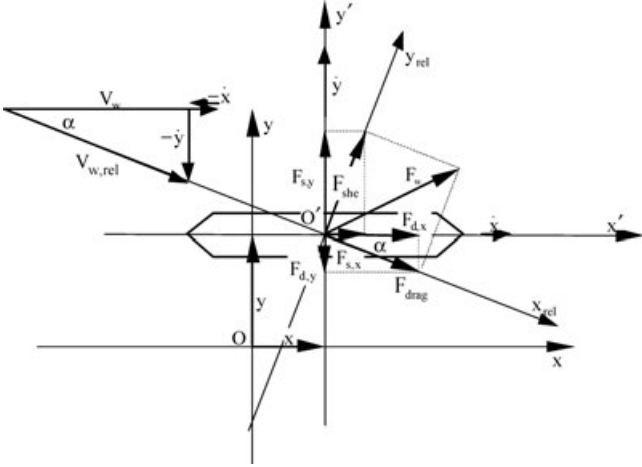


Fig. 1. Deck cross section and aerodynamic forces.

velocity $\vec{V}_{w,rel(t-\Delta t),x}$ at time $t - \Delta t$. The modulus of $\vec{F}_{she(t),x}$ is evaluated at time t as a function of the values at time $t - \Delta t$, according to the following expressions (D'Asdia and Noè, 1998):

$$F_{she(t),x} = \frac{1}{2} \rho H C_{L-she} V_{w,rel(t),x}^2 \times \sin(\omega_{she(t),x} t + \varphi_{she(t),x}) \quad (2)$$

$$V_{w,rel(t),x} = \sqrt{V_{w(t-\Delta t),x}^2 + \dot{y}_{(t-\Delta t),x}^2} \quad (3)$$

$$\alpha_{(t)} = \tan^{-1}(\dot{y}_{(t-\Delta t)}/V_{w(t-\Delta t)}) \quad (4)$$

$$\omega_{St(t),x} = 2\pi St \frac{V_{w,rel(t),x}}{H}; \quad \tilde{\omega}_{(t),x} = \frac{\pi}{\tau_{n,(y_x=\bar{y}_x)} - \tau_{n+1,(y_x=\bar{y}_x)}} \quad (5, 6)$$

$$\begin{cases} \omega_{she(t),x} = \omega_{St(t),x} & \text{if } \frac{\omega_{St(t),x}}{\tilde{\omega}_{(t),x}} < \Omega_L \text{ or } \frac{\omega_{St(t),x}}{\tilde{\omega}_{(t),x}} > \Omega_U \\ \omega_{she(t),x} = \omega_{(t),x} & \text{if } \Omega_L \leq \frac{\omega_{St(t),x}}{\tilde{\omega}_{(t),x}} \leq \Omega_U \end{cases} \quad (7)$$

$$\varphi_{she(t),x} = [\omega_{she(t-\Delta t),x} - \omega_{she(t),x}]t + \varphi_{she(t-\Delta t),x} \quad (8)$$

The term $y_{t,x}$ denotes the across-wind deck displacement of the generic cross section (vertical), ρ the air density, $\bar{y}_x$ is the mean across-wind displacement of the section, and the dot symbol denotes the time-derivative. The quantity $\tilde{\omega}_{(t),x}$ is the estimated value of the instantaneous circular frequency of the across-wind oscillation

due to vortex shedding, derived by computing the difference between $\tau_{n,(y_x-\bar{y}_x)}$, $\tau_{n+1,(y_x-\bar{y}_x)}$, the time instants corresponding to the last two zero-crossings of the function $y_x - \bar{y}_x$, representing the across-wind dynamic displacement component estimated from the numerical analyses.

This is a unique feature of the model because the synchronization condition is not pre-calculated “*a priori*” in terms of a set of initial natural frequencies of the structure, but is determined, instant by instant, as a function of the local “instantaneous” oscillation frequency, numerically calculated at each node of the finite-element model of the structure.

The modulus of the component $\vec{F}_{drag}$ due to the drag and parallel to the relative wind velocity, is evaluated at time t by means of the standard definition of aerodynamic drag, that is,

$$F_{drag(t),x} = \frac{1}{2} \rho B C_D V_{w,rel(t),x}^2 \quad (9)$$

where C_D is the static drag coefficient of the cross section. Vortex-shedding-related coefficient C_{L-she} and St (Strouhal number) in Equations (2), (5), and (6), applicable to cross sections belonging to the deck or to the main cable, are normalized with respect to H , deck height, or D_{eq} , cable diameter, respectively. A fully correlated spatial representation of aeroelastic forces at lock-in is assumed in the vortex-shedding model.

2.2 Simulation of aerodynamic loading, not associated with vortex shedding

Aerodynamic loading derived from other sources and not directly related to vortex shedding, was simulated through quasi-steady formulation (e.g., Simiu and Scanlan, 1996), such as turbulence-induced or motion-induced-excitation (MIE) forces (e.g., Naudascher and Rockwell, 1994).

In relation to the deck, this formulation is based on the definition of drag, lift, and moment forces per unit length as a function of the angle of attack of the wind, which includes the effects of the dynamic fluid-structure interaction (i.e., relative velocity component as a function of time t). The model employs aerodynamic coefficients dependent on the local angle of attack (α), which are related to drag, lift, and moment per unit length and defined as $C_D(\alpha)$, $C_L(\alpha)$, and $C_M(\alpha)$, respectively [please, note Equation (9) for comparison]. These are normalized with respect to the deck width B . A detailed description of the model can be found in Caracoglia (2001) and Sepe et al. (2000).

To this respect, the model neglects the effects of non-stationarity in the MIE forces not associated with vortex

shedding. Nevertheless, simulations conducted by the authors on the same bridge models (Caracoglia, 2001; Caracoglia et al., 2000), as described later in Section 4, performed for a set of distinct wind conditions (including, for example, those in the proximity of the aeroelastic instability threshold) confirmed that the use of static coefficients was acceptable, and this observation was used during the preparation of the current study. In any case, these effects should possibly be included in future extensions of the model, for example, based on the representation of aeroelastic forces through indicial functions (e.g., Caracoglia and Jones, 2003).

The proposed model also enables the evaluation of the wind loading on the two towers. Nevertheless, although the model was readily available, it was not included in the examples described in this article because the main goal of this study was related to the analysis of the potential interaction of phenomena involving the vibration of the suspended portions of the bridge.

A methodology for the assessment of the effects associated with the oncoming wind turbulence on the vortex-shedding response of long-span bridges (D'Asdia et al., 1996) and an algorithm for the numerical generation of turbulent wind fields (Viskovic, 1996; D'Asdia and Viskovic, 1996) are available. However, the description of the algorithm was omitted in this article. Because the main purpose of the study was the discussion on the dynamic effects associated with vortex shedding, these algorithms were in fact not directly employed in the numerical simulations in Section 4.

Even though a quasi-steady representation of the aerodynamic forces not associated with vortex shedding is employed, the proposed nonlinear vortex-shedding model can be considered as adequate for the purposes of the evaluation of the structural response because the effects of flow unsteadiness, primarily related to the presence of the vortices, are incorporated in the numerical procedure (see Section 2.1). Moreover, numerical simulations are restricted in this study to the range of low to medium wind velocities with low turbulence in the oncoming flow, in which vortex-induced-vibration (VIV) effects will be predominant, whereas other MIE mechanisms will be not engaged. Therefore, simulations in Section 4 must be interpreted as the effects of the combination of an average aerodynamic loading with those induced by VIV.

3 THE FINITE-ELEMENT COMPUTER MODEL

The development of time-domain models is connected to the possibility of considering the nonlinearity of the system, which cannot be directly analyzed in the

frequency-domain. Indeed, this aspect becomes important in the case of suspension bridges. The nature and the complexity of the structure can also influence the selection of the numerical procedures (e.g., an appropriate integration scheme must be considered, a large number of degrees of freedom is usually required, etc.).

There are fundamentally three problems that need to be considered: the nonlinear behavior of the structure, the description of the wind loading (Section 2), and the overall dimension of the numerical problem. Although the third aspect is seldom of concern in the recent years, the first aspect is of relevance. In particular, nonlinear behavior is mostly concentrated in the suspension system (e.g., stiffness accounting for the geometrical effects of the main cable in a suspension bridge) and must be carefully considered.

The computer model for vortex-shedding loads, described in Section 2, was employed in conjunction with a nonlinear finite-element model ("Tenso"), originally developed by D'Asdia and co-authors (2003) for the analysis of nonlinear structures, and more recently extended to include the wind-induced response of long-span bridges (e.g., Sepe et al., 2000; Caracoglia et al., 2000).

The current version of the computer model allows for finite-element analysis in the time domain of suspension bridges and other large structures, simulated by one-dimensional elements, taking into account the geometrical nonlinearities of the elements. The vortex-shedding-induced loads, that are of main concern in this article, were modeled as described in Section 2.1, while the buffeting forces acting on deck, towers, and cables are evaluated through static coefficients (Section 2.2); the effective angle of attack is evaluated at each time instant by combining the effects of the wind speed, bridge deformation, and motion in accordance with the quasi-stationary method.

Any configuration of the wind field (mean value, correlation functions, and turbulence) can be introduced to evaluate the wind speed on each element (deck, hangers, cables, towers, etc.). The "Tenso" framework allows for elastic nonlinear static and step-by-step dynamic analyses of structural types, which can be discretized through monodimensional elements (either cable or beam type).

A three-step numerical procedure was employed for the implementation of the computer model. The first one allows for modal analysis and natural frequency extraction. In the second one, the static or dynamic nonlinear time-history integrations are carried out. The solution to the elastic problem is performed in terms of total Lagrangean coordinates. Cable-type finite elements are of parabolic or rectilinear shape, while deck elements are beam-type (Timoshenko).

Static problems can be solved by a second-order analysis, by means of an iterative procedure involving a succession of load steps, during which geometrical stiffness of all the elements is continuously updated. The dynamic analysis is performed through a step-by-step integration of the motion equations, using Newmark's procedure, therefore ensuring absolute convergence. The time step Δt (Section 2.1), preliminarily selected at the beginning of each simulation to ensure general accuracy, is automatically recalibrated during consecutive time steps in the dynamic analyses to account for potential changes in the structural properties due to nonlinear response (e.g., changes in the coefficients of the stiffness matrix), which may affect the numerical solution.

The mechanical damping is assumed in accordance with Rayleigh's method; an accurate calibration of the damping coefficients of the method will be of relevance for the investigated cases because compatibility of the numerically generated damping levels with existing structures must be enforced.

4 APPLICATION EXAMPLES

The global aerodynamic and aeroelastic model described in Sections 2 and 3 was applied to two numerical examples, already investigated in previous studies in relation to the problem of aeroelastic instability (Sepe et al., 2000). In this section, the attention is paid to the effects due to vortex shedding, as described by the model discussed in Section 2.1 and specifically tailored in this article to the analysis of bluff bodies such as bridge decks. In particular, both examples correspond to the same structural model, that is, the 1992 design of the Messina Bridge, characterized by a three-box streamlined cross section. However, in the second case the original deck was replaced by a scaled struc-

tural model of a single-box girder with inferior aerodynamic performance to assess the amplification due to vortex-induced effects. For this purpose, the simulation of the VIV and MIE forces on a single-box girder was based on the aerodynamic characteristics of the Humber Bridge built in the United Kingdom (e.g., Brownjohn et al., 1994), which was designed for a similar structure with a much shorter span (about 1400 m). In particular, the model of the 3,000-m simulated structure does not correspond to the real Humber Bridge, which has completely different structural characteristics and span, and which did not exhibit in reality large VIV (Bocciolone et al., 1992; Diana et al., 1992; Brownjohn et al., 1994; Brownjohn et al., 1995), as later described in Section 4.2.

Results are also based on a preliminary study which was conducted on a restricted set of cases by partially neglecting nonlinear loading on some cable/deck elements (Caracoglia et al., 2000).

4.1 A first example: the 1992 design of the Messina Bridge

The first example is relative to the 1992 design of the bridge over the Messina Strait (Italy), a road and railroad bridge with central span of 3,300 m, with tower height 374 m and mean deck elevation of 68 m above the sea level. The cross-section consists of three parallel boxes, linked to the suspension system by means of transversal elements distant 30 m from each other.

In Table 1 the main aerodynamic characteristics are summarized. In the model of vortex shedding described in Section 2, the values $\Omega_L = 1.00$ and $\Omega_U = 1.20$ were assumed [Equation (7)], according to previous studies (D'Asdia and Noè, 1998). As concerns St and C_{L-she} (Strouhal number and vortex-shedding transverse force

Table 1
Aerodynamic characteristics of the deck designed for the Messina Bridge
(Example 1)

Bridge deck	Main cable	Cross section
$C_D(0) = 0.090$		
$C_L(0) = -0.055$	$D_{eq} = 1.6$ m	
$C_M(0) = 0.020$	$St = 0.21$	
$B = 62$ m	$C_{L-she} = 0.42$	
$H = 6.0$ m (equiv.)		
$St = 0.083$		
$C_{L-she} = 0.50$		

Note: Drag, lift, and moment coefficients (C_D , C_L , C_M) are normalized with respect to the deck width B . Vortex-shedding-related coefficient C_{L-she} and St (Strouhal number) are normalized with respect to H (deck height) or D_{eq} (cable diameter), for the deck and the cable respectively.

coefficient, respectively), the indications of the Eurocode on “Wind actions” (EC-1, 2002) were also considered. Moreover, the former quantity seems compatible with the second lock-in regime measured in the wind tunnel by Diana et al. (2006) on a 1:60 scale model of the Messina Bridge (reduced velocity $V^* = V_w/fB = 1.0$), from which a $St = fH/V_w \approx 0.09$ can be estimated (additional wind tunnel experiments would be necessary to confirm the presence of this regime).

Experimental studies conducted on circular cylinders and full-scale data (D’Asdia and Noè, 1998) suggest that estimation of C_{L-she} (lift shedding coefficient) and the interval of reduced velocities V_w/fB (i.e., Ω_L and Ω_U in the model), for which lock-in is observed, can be performed in a wind tunnel by simulating lock-in conditions on a single-degree-of-freedom reduced-scale model, with width B and oscillating at frequency f . Although influence of the vibration amplitude on the magnitude of the experimentally derived coefficients cannot be excluded due to the inherently nonlinear nature of the lock-in phenomenon, numerical simulations confirm that self-limiting vibration regimes are accurately reproduced through the proposed nonlinear model with constant aerodynamic coefficients (D’Asdia and Noè, 1998).

Moreover, it must be noted that in the case of a complex bridge deck, such as a multi-box cross section with intermediate openings (Table 1), the nature of the vortices responsible for the mechanism generating the vibration may not be associated with the classical “von Karman type.” For example, other mechanisms for the generation of vortices (e.g., “impinging shear layers”) have been suggested for rectangular prisms depending on the width-to-depth ratio (Matsumoto, 1999; Nau-dascher and Rockwell, 1994). Recent wind tunnel experiments conducted on oscillating models of the three-box Messina Bridge deck (Diana et al., 2006) tend to confirm the presence of a complex mechanism of excitation, in which more than one lock-in regime was observed. This fact therefore reveals an inherent complexity of the aeroelastic loading, which may require careful consideration, beyond the original scope of this study. Nevertheless, the proposed computer model for the simulation of VIV can be adapted to other vortex regimes, provided that distinct sets of parameters St and C_{L-she} can be recorded in a wind tunnel test, that is, observed during the tests along with the presence of distinct reduced velocity intervals for susceptibility to lock-in.

The wind speed vertical profile was modeled in accordance with the logarithmic law (Simiu and Scanlan, 1996), referred to the 10-minute average value V_d at the deck level. The range of variation of this value was considered between 4 and 22 m/s, to investigate the possibil-

ity of synchronization (lock-in) both on the main cable and the deck. The lower limit of this wind speed interval corresponds to a vortex shedding from the main cables with $V_c = 6$ m/s (10-minute average) and a frequency around 0.6 Hz, close to the frequency of a global vertical mode with wavelength $\lambda_{vert} = 450$ m. The upper limit can be associated with vortex shedding from the deck with a frequency of 0.3 Hz, close to the frequency of a vertical mode with $\lambda_{vert} = 700$ m.

To simplify the evaluation of the aerodynamic forces, each pair of suspension cables (two cables on each side) was replaced, as a first approximation, by a single cable with diameter of 1.6 m, with equivalent area. In particular, aerodynamic loads induced by vortex shedding on the elements of the structure for wind speeds V_d lower than 15 to 17 m/s are not relevant for the structural response and they are absolutely negligible for $V_d = 4$ m/s ($V_c = 6$ m/s), corresponding to lock-in on the main cables. However, in the case of $V_d = 22$ m/s some interesting results were found (lock-in phenomenon on the deck). This case only is described here in detail.

For this value of wind speed ($V_d = 22$ m/s) the influence of the mechanical damping is also discussed. Because the structural modal damping coefficient ξ (percentage with respect to critical) depends on the frequency of the considered mode (Rayleigh method), the parameters are usually chosen to ensure reasonable values for ξ (sufficiently close to the mean value $\bar{\xi}$) in the frequency range corresponding to the first modes of the bridge, that is, between 0.03 and 0.10 Hz. For higher modes with a period of about 4 seconds, such as those involved in this part of the analysis, the structural damping $\xi|_{T=4s}$ is therefore higher. The “aerodynamic damping” is also included in the model (quasi-stationary approach for the effective angle of attack).

The most interesting results were found for mean structural damping ratio $\bar{\xi}$ equal to 0.4% and 0.2%, that is, for $\xi|_{T=4s} = 0.7\%$ and $\xi|_{T=4s} = 0.4\%$, respectively. In particular, for $\xi|_{T=4s} = 0.4\%$ a maximum vertical displacement of 0.13 m was observed in the mid-span section of the deck (Figure 2a), while for $\xi|_{T=4s} = 0.7\%$ the same quantity is considerably lower (Figure 2b).

For $V_d = 22$ m/s and $\xi|_{T=4s} = 0.4\%$, the maximum amplitude of the vertical displacement v_{z-rel} at the mid-span section, with respect to the quasi-steady deflection due to static loads (v_{z-mean}), is equal to 0.08 m (Figure 3a). The corresponding vertical accelerations are about 0.26 m/s^2 and, therefore, almost irrelevant. The oscillation period is equal to 3.5 seconds, close to that of a vertical mode with wavelength of about 740 m (Figure 3b).

It must be observed that these results, obtained through the proposed model for the Messina Strait Bridge, are very close to the experimental values

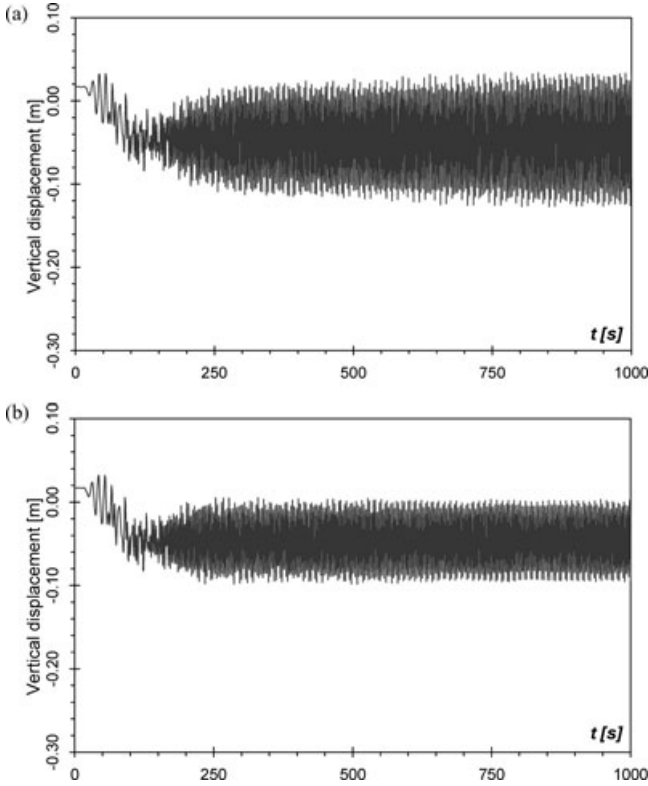


Fig. 2. Vertical displacement of the mid-span deck section of the Example 1, with $V_d = 22$ m/s; (a) $\xi|_{T=4s} = 0.4\%$, (b) $\xi|_{T=4s} = 0.7\%$.

indicated by Diana et al. (1995) on the full-bridge aeroelastic model of the structure, in particular for wind speeds at deck level between 10 and 20 m/s (vertical acceleration of the deck).

For $V_d = 22$ m/s, the wind speed at the cable height corresponds to vortex-shedding frequencies of approximately 20 Hz (Strouhal Law) and aerodynamic forces of 0.3 kN/m, negligible in comparison with those associated with lock-in from the deck (about 2 kN/m). Relative displacement between main cables and the deck in the vertical direction is almost absent.

For vortex-induced excitation corresponding to wind-speed levels close to the flutter threshold for this bridge (U_d between 60 and 80 m/s), the interaction between dynamic loading on the main cables and aerodynamic forces originating from the deck can be excluded because the shedding frequencies for main cables would be of the order of 8 to 10 Hz in accordance with the Strouhal relationship, that is, considerably larger than the flutter frequency (estimated as 0.07 Hz for this bridge), which is predominant in this interval of wind speeds. These observations were also confirmed by numerical results obtained through the application of the time-domain simulation framework described in this article but are not

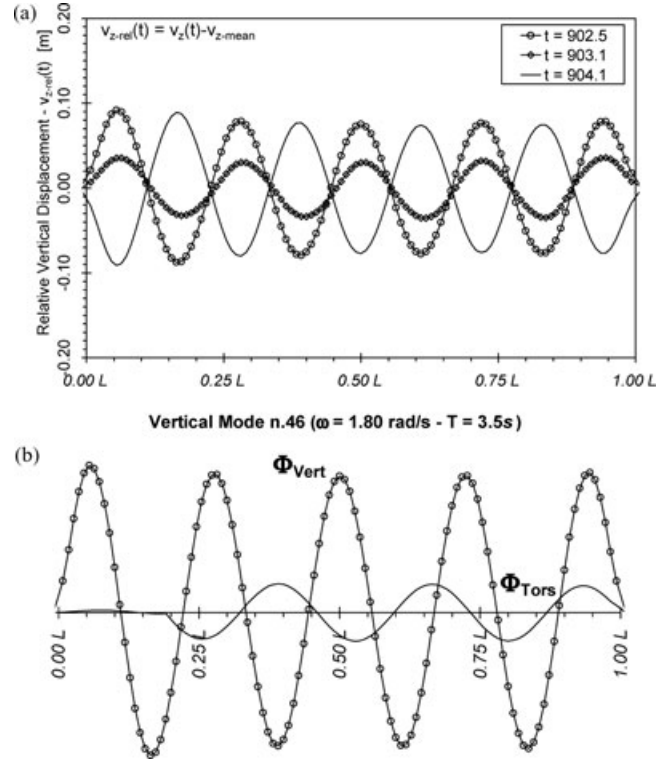


Fig. 3. Lock-in vibration simulations of the Messina Bridge (Example 1) with $V_d = 22$ m/s and $\xi|_{T=4s} = 0.4\%$. (a) Evolution of the relative vertical displacement (v_{z-rel}) of the deck at three different time t ; (b) Flexural mode shape, involved in the oscillation.

reported here. In particular, simulations tend to show a small delay in the flutter onset if dynamic loading associated with vortex shedding from the main cables is included.

4.2 A second example: single-box deck section (Humber-bridge type)

In this second example, the same static and geometrical characteristics of the 1992 design of the Messina Strait Bridge (Section 4.1) were considered. Without modification to the deck mass the system was replaced by an equivalent single-box girder of width $B = 67.26$ m, with aerodynamic characteristics similar to the Humber Bridge (e.g., Brownjohn et al., 1994). From the comparison of Table 2 with Table 1, the reduction of aerodynamic performance is evident and due to the fact that the second cross section is less streamlined than the original multi-box.

The choice of this fictitious cross section was motivated by the need for a better interpretation of the role played by vortex shedding from the deck on the global response.

Table 2
Aerodynamic characteristics of the single-box deck (Example 2)

Bridge deck	Main cable	Cross section
$C_D(0) = 0.090$		 (Humber-Type)
$C_L(0) = -0.255$	$D_{eq} = 1.6$ m	
$C_M(0) = 0.045$	$St = 0.21$	
$B = 67.26$ m	$C_{L-she} = 0.42$	
$H = 10.67$ m		
$St = 0.130$		
$C_{L-she} = 0.71$		

Note: Drag, lift, and moment coefficients (C_D , C_L , C_M) are normalized with respect to the deck width B . Vortex-shedding-related coefficient C_{L-she} and St (Strouhal number) are normalized with respect to H (deck height) or D_{eq} (cable diameter), for the deck and the cable respectively.

It must be noted, however, that this second example is not related to the actual Humber Bridge, for which experimental and full-scale results are available (Bocciolone et al., 1992; Diana et al., 1992; Brownjohn et al., 1994; Brownjohn et al., 1995). The simulated structure is different, that is, a suspension bridge with central span two-and-a-half times longer than that of the Humber Bridge (1440 m). In particular, geometry and physical characteristics of the simulated deck and cables were also calibrated to be representative of a realistic 3,000 m structure with mass and flexural/torsional stiffness compatible with the presence of a single-box girder of large dimensions (twice the width of the original deck). As a result, fundamental periods of the simulated structure are of the order of 30 seconds (first lateral mode) to more than 7 seconds (first torsional mode) in contrast with the real structure [e.g., approximately 3 seconds were indicated for the fundamental torsional mode in the experimental study by Brownjohn et al. (1994)]. Therefore, differences between the simulations on the modeled bridge, presented in this study, and full-scale results on the actual bridge must be expected and direct comparison does not seem possible.

In the numerical simulations, wind speed V_d at the deck level varying between 10 and 23 m/s was considered. Similarly, to the previous case, two different values of structural damping were considered.

In the case $V_d = 23$ m/s, for $\xi|_{T=4s} = 0.4\%$ and 0.7% , Figure 4 depicts the vertical displacements of the mid-span cross section of the deck, corresponding to an antinode for the natural mode excited by vortex shedding (frequency 0.25 Hz). The value of V_d was in fact carefully selected to enhance the potential interaction with a higher vertical mode in the frequency interval between 0.24 and 0.27 Hz. Maximum displacements of the order of 0.50 m were observed for both values of damping considered: the oscillation reduction due to an increment in the structural damping was less significant in this case.

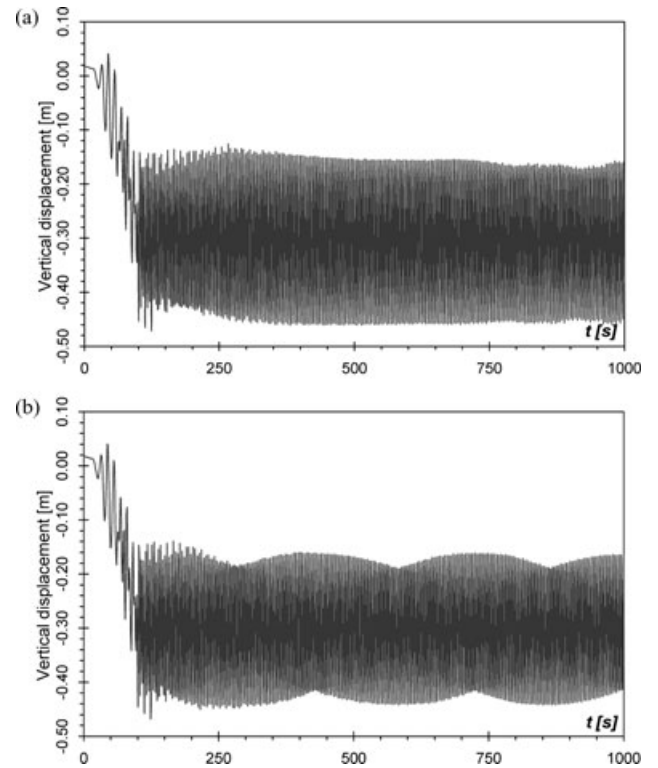


Fig. 4. Vertical displacement of the mid-span deck section of Example 2 (single-box deck section) with $V_d = 23$ m/s; (a) $\xi|_{T=4s} = 0.4\%$, (b) $\xi|_{T=4s} = 0.7\%$.

As a consequence of the synchronization, the amplitude of the vertical relative displacement v_{z-rel} is equal to 0.16 m for $\xi|_{T=4s} = 0.4\%$ (Figure 5a). Despite the aerodynamic characteristics of the deck, this value is not large but not negligible. In addition, it was obtained for a relatively low wind speed regime, 50% lower than the critical flutter speed (54 m/s) and often expected during a time period corresponding to the typical lifetime of a bridge structure. The oscillation frequency is equal to 0.25 Hz, close to the frequency of a seven-half-wave

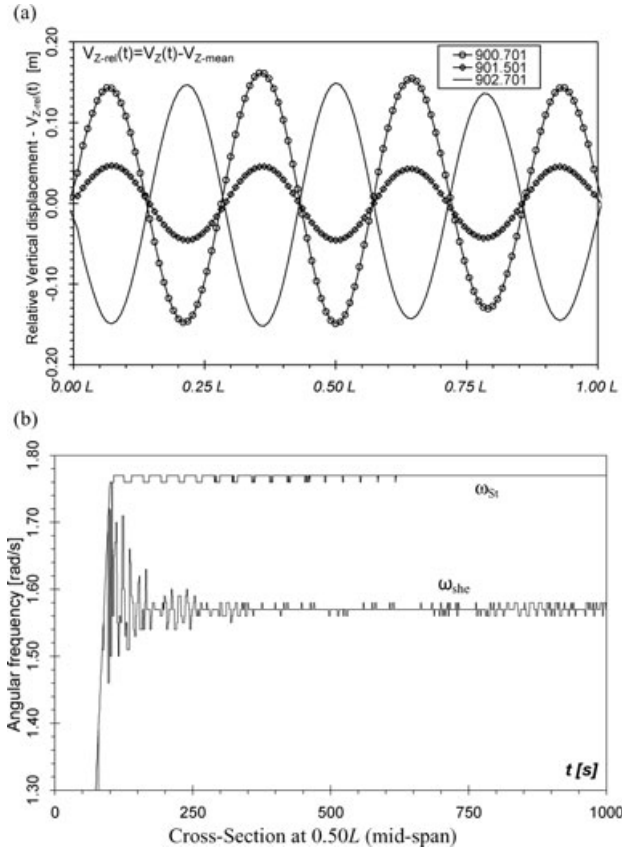


Fig. 5. Lock-in simulations of Example 2 (single-box deck section) with $V_d = 23$ m/s and $\xi|_{T=4s} = 0.4\%$. (a) Evolution of the relative vertical displacement (v_{z-rel}) of the deck at three different time instants t ; (b) evolution of the lock-in force frequency (ω_{she}) in comparison with the theoretical value ω_{St} (Strouhal relationship).

mode ($\lambda_{vert} = 940$ m—Figure 5a), where a significant torsional component is also present.

Figure 5b shows the time history of the angular frequency ω_{she} of the periodic vertical force at the mid-span section of the deck, computed by means of the procedure summarized in Equations (3)–(8), while ω_{St} represents the angular frequency calculated by the Strouhal law. It is evident that ω_{she} almost always deviates from the theoretical value ω_{St} due to the dominant synchronization (lock-in) on the deck.

Numerical investigation also showed the presence of significant relative displacements between the main cables and the deck, corresponding to cyclic variations in the tension of the hangers of about 5–10% with respect to the mean value for the cross sections close to $0.25L$. Although not very large, these values should be monitored in relation to potential long-term fatigue effects on the hangers.

As concerns the travel comfort, the ISO-2631 code (ISO Guidelines for User Proficiency), was considered,

as suggested by Larsen and Jacobsen (1992) for the Great Belt, assuming a tolerance limit for the RMS acceleration of 1.5 m/s^2 for oscillations with frequency lower than 1 Hz. Therefore, in both cases and in particular for the second example described in this section vertical deck accelerations (around 0.40 m/s^2 , RMS 0.28 m/s^2) are below this critical threshold. In any case, the degree of discomfort can be located in the range of “annoying disturbances” (Simiu and Scanlan, 1996), with accelerations between $0.015g$ and $0.05g$ (g : gravity acceleration), potentially uncomfortable for the users of a railroad bridge.

5 DISCUSSION ON THE EFFECTS OF VORTEX SHEDDING ORIGINATING FROM THE VIBRATION OF THE MAIN CABLES

The computer model and its unique features enable the simulation of vortex-shedding loading on all the elements of a line-like structure and for any cross-sectional geometry. These include not only the main deck but also the main cables and hangers. Nevertheless, the primary objective of this study was the numerical simulation of VIV originating from the bridge deck. For this particular case, that is, at the low to medium velocity regimes corresponding to VIV from the main deck, specific problems and potential issues associated with vortex shedding from the main cables or the hangers are not relevant.

In particular, global modes of the bridge, for which main cable oscillation is conceivable and vortex-shedding effects may be of relevance, are usually associated with large modal masses, therefore resulting in large mass damping parameter (i.e., Scruton number; Simiu and Scanlan, 1996). Preliminary simulations conducted on the model of the 1992 Messina Bridge confirmed that wind velocities, for which this excitation mechanism is indeed possible, are usually extremely low, and therefore associated with negligible available input energy from the wind, which is clearly incapable of exciting such a large system.

At higher wind velocities, localized excitation in the higher modes of the main cables is possible. This is a condition that may be recorded at extremely high frequency for the main cable, potentially dangerous for the safety of the cable (e.g., fatigue). Nevertheless, in this case the presence of large turbulence in the oncoming flow would have a beneficial influence by reducing the spatial coherence of the vortices responsible for the onset of the vibration (Simiu and Scanlan, 1996).

Finally, although local vibration of individual hangers is possible, as reported in the literature (e.g., Cigada et al., 1997), this is often related to local wake-induced

effects, which are not directly contemplated by the current computer model.

6 CONCLUDING REMARKS

This article describes the development, simulation, and validation of a new computer procedure for wind loading response simulation on long-span bridges in the time domain. This model, originally developed for aeroelastic instability and buffeting analyses with loading exclusively concentrated at the deck level, was re-formulated in this research to also include other aerodynamic effects, such as those related to vortex shedding.

The computer model, employed to simulate the dynamic bridge response under different combinations of aerodynamic loading, can simultaneously account for aerodynamic and aeroelastic forces on all elements of the bridge, such as deck, cables, and towers.

The model used for the simulation of vortex-induced loading is based on an existing model, already developed for the dynamic response of chimneys, that is, very slender vertical structures. In the case of chimneys, the validation of the model was performed through experimental data derived from the literature, wind tunnel tests, and full-scale experimentation performed by the authors (D'Asdia and Noè, 1998). The combination of this model with a nonlinear finite-element computer model for structural analysis in the range of large-amplitude oscillations is novel and unique.

Indeed, the general question on the superposition of fluid-dynamic forces originating from turbulence, vortex-induced-vibration, and other motion-induced-excitation phenomena is research question, which is still partially open and was considered beyond the original objective of the study. Superposition of these excitation mechanisms is accepted in the literature provided that the vibration amplitudes are low. However, the simulations presented in this study confirm that, even in the presence of all excitation sources in the modeling (VIV and MIE), oscillation is dominated by the effects of vortex shedding and the contribution of other loading mechanisms to the dynamic response is often negligible. The authors also believe therefore that the combination of the turbulence-induced forces and vortex-shedding loading is possible within the existing computer modeling framework, provided that the hypothesis of independence between the two excitation sources is considered (as usually acceptable in most applications).

In this study, emphasis was given to vortex-shedding excitation. The direct estimation of the response in the time domain was possible through a simulation algorithm which extends to bridge deck sections a numeri-

cal model proposed for circular cross sections only, and which only requires few fluid-dynamics parameters to be specified or derived from basic section-model tests in a wind tunnel. Most physical quantities of the structure that directly influence the onset of the vortex-induced vibrations (such as the influence of geometric nonlinearity and the effects of the initial angle of incidence of the wind), are in fact automatically detected by the computer procedure during the simulations.

Two numerical examples of bridges with single or multiple closed-box girders were employed to numerically validate the proposed algorithm. Analyses have shown that the combination of aeroelastic and aerodynamic loading on the main cables and the deck must be included in the estimate of structural response, especially during vibration regimes dominated by vortex shedding; a correct assessment of the dynamic response may in fact become of particular relevance during the lifetime of the structure for fatigue problems and for the reduction of user discomfort. These observations are particularly useful to the bridge designer: specific simulations need to be considered to include potential interaction among different loading mechanisms (including vortex shedding).

Nevertheless, the interpretation of the results discussed in this article must be carefully interpreted in relation to the reliability of the proposed model because numerical analyses are available at the moment only. Further investigation, including the comparison with either laboratory experiments or full-scale data, is therefore required for an accurate interpretation of the phenomenon and its relevance to design.

Future developments of the research will possibly include the following aspects.

1. Experimental validation of the computer model (either in the wind tunnel or full-scale).
2. Simulation of the effects of turbulence in the oncoming flow on VIV, not considered at the current stage.
3. The effects of span-wise coherence in the upstream and in the wake as a means of reducing or enhancing the action of vortex shedding. Passive (or active) aerodynamic devices for the reduction of the coherence in the structure of the vortices have in fact been indicated as a viable option for suppressing VIV (e.g., Larsen et al., 2000). Nevertheless, the use of such devices could produce a potential increment in the drag forces that might be detrimental during other events such as extreme wind situations, and therefore experimental analysis would be required to better quantify such effects. In any case, the computer model developed in this study would be capable of simulating

the presence of such localized devices, for example, by suitable modification of the parameters describing the nonlinear vortex-shedding model.

4. Nonlinear effects of stay cables as remedial devices due to an increment of aerodynamic damping associated with the presence of local vibrations.
5. The efficiency of tuned mass dampers on suppressing vortex-shedding vibrations; this is still a controversial issue. Although some researchers indicate that they work better when a nonlinear force model is used (e.g., Strømme and Hjørth-Hansen, 2001; Livesey and Larose, 1995), some situations may arise, in which they seem to be ineffective (Owen et al., 1996).

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NOMENCLATURE

The following symbols have been used in this article:

- $C_D(\alpha)$ static drag coefficient per unit length, angle of attack α
- $C_L(\alpha)$ static lift coefficient (independent of vortex shedding) per unit length, angle of attack α
- C_{L-she} vortex-shedding force coefficient
- $C_M(\alpha)$ static moment coefficient per unit length, angle of attack α
- B deck width
- f natural frequency, Hz
- $\vec{F}_{drag(t),x}$ time-dependent drag force vector component per unit length (Figure 1), cross section x
- $\vec{F}_{she(t),x}$ time-dependent vortex-shedding force vector per unit length (Figure 1), cross section x
- $\vec{F}_{w(t),x}$ time-dependent total force vector per unit length (Figure 1), cross section x
- g acceleration of gravity
- H deck height
- St Strouhal number fH/V_w
- t time variable
- T structural mode period
- Δt time step (numerical algorithm)
- V_{z-mean} across-wind deflection due to static loads of the bridge deck (quasi-steady)
- V_{z-rel} maximum relative amplitude of across-wind vibration of the bridge deck
- V^* reduced velocity (V_w/fB)
- V_c reference 10-minute wind velocity at the main cable level (log-law)
- V_d reference 10-minute wind velocity at the deck level (log-law)
- $V_{w,x}$ cross-flow velocity, cross section x

$\vec{V}_{w,rel(t),x}$	relative velocity (fluid–structure interaction) at time t , cross section x	$\xi _{T=4s}$	structural modal damping for modes with $T = 4s$, evaluated through Rayleigh method
x	longitudinal abscissa along the bridge axis	ρ	air density
$y_{(t-\Delta t),x}$	vertical bridge displacement (generic cross section) at time $t - \Delta t$	$\tau_{n,(y_x - \bar{y}_x)}$	time instant corresponding to the n th zero crossing of the function $y_x - \bar{y}_x$
$y_{(t),x}$	vertical bridge displacement (generic cross section) at time t	$\tilde{\omega}_{(t),x}$	estimated circular frequency of transverse bridge oscillation, time t , cross section x
$\alpha_{(t),x}$	instantaneous angle of attack of the wind loading, time t , cross section x	$\omega_{she(t),x}$	instantaneous vortex-shedding circular frequency, time t , cross section x
$\varphi_{she(t),x}$	instantaneous vortex-shedding phase lag, time t , cross section x	$\omega_{St(t),x}$	instantaneous Strouhal-relationship circular frequency, time t , cross section x
λ_{vert}	wavelength corresponding to a vertical bending mode of the deck $\bar{\xi}$ mean structural modal damping over different periods T (Rayleigh method)	Ω_L	lower bound of the lock-in interval dimensionless frequency
		Ω_U	upper bound of the lock-in interval dimensionless frequency